

hacking HikVision IP cameras



agenda

- ~5m Introduction
- ~30m Technical Deep Dive
- ~5m Questions

whoami

[Name] Hrant
[Interests] Hardware Hacking, Reverse Engineering, Exploits...

Talks:

[ArmSec 2023] Visual Reverse Engineering
[ArmSec 2024] BIOS Malwares
[ArmSec 2024] Reverse Engineering The MacOS Sandbox
[CyberGen 2025] Reverse Engineering DJI Drones, Hardware & Software
[ArmSec 2025] Hacking HikVision IP Cameras

links

[Repository] <https://github.com/0x0000z3r0/hikvision-re>

Contact:

[Email] hrant.tadevosyan@protonmail.com

[LinkedIn] <https://www.linkedin.com/in/hrant-tadevosyan>

[Discord] 0x0000z3r0 (Z3R0)

motivation

- [+] Curiosity
- [+] Security
- [+] Modding

motivation

[+] Curiosity

[-] Security

 [*] Used everywhere in **Armenia**: banks, buildings, buses, etc.

 [*] Privacy concerns: **not** open hardware and/or source

[+] Modding

motivation

- [+] Curiosity
- [+] Security
- [-] Modding
 - [*] Extend/patch the firmware or hardware
 - [*] Fix the problems yourself

trophies

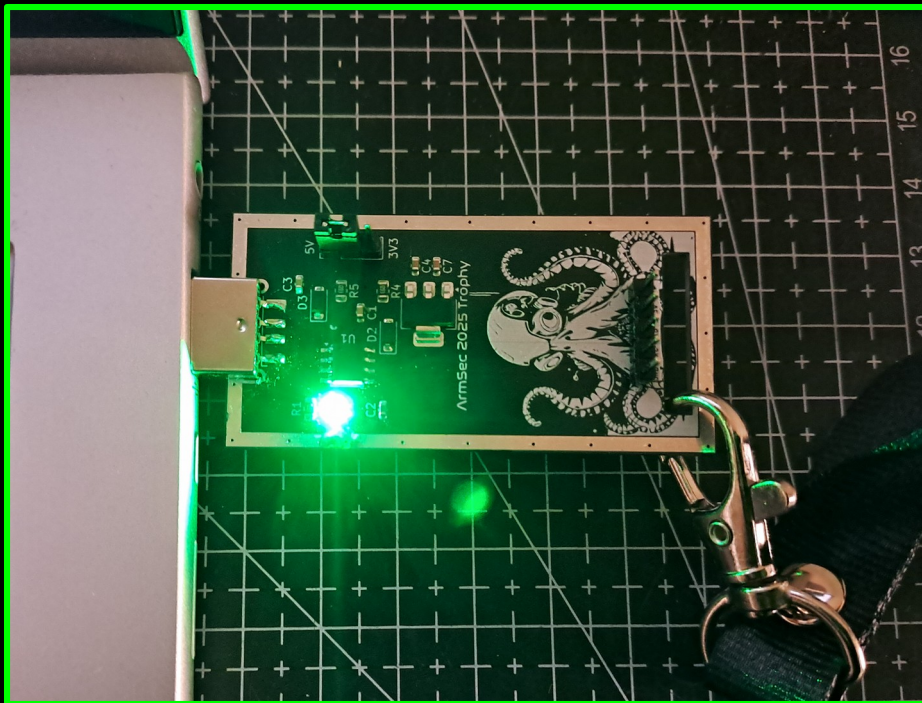
[Badges] for attentive listeners

[Difficulty] easy+



trophies (badges)

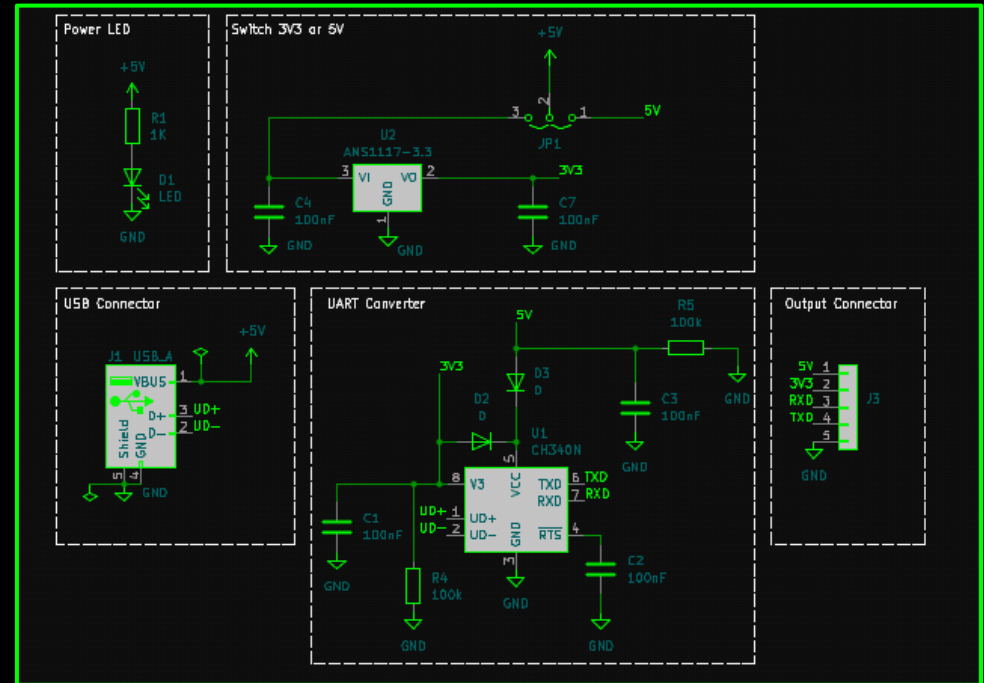
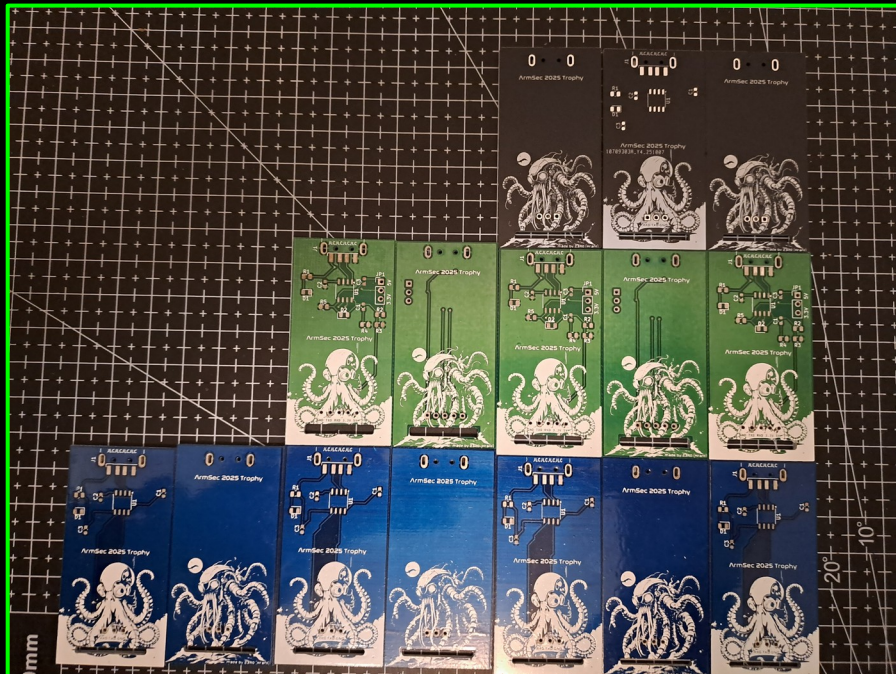
- [*] uart to usb converter
- [*] custom board design with ArmSec logo
- [*] was designed by a software person... bugs ahead



trophies (prototypes)

[*] parts need to be sourced & soldered manually

[*] btw we have a local PCB manufacturer, called **HyePeak**



clarification

[hacking] hardware hacking

[cameras] only one

...

[enviornment] casual, not a lab

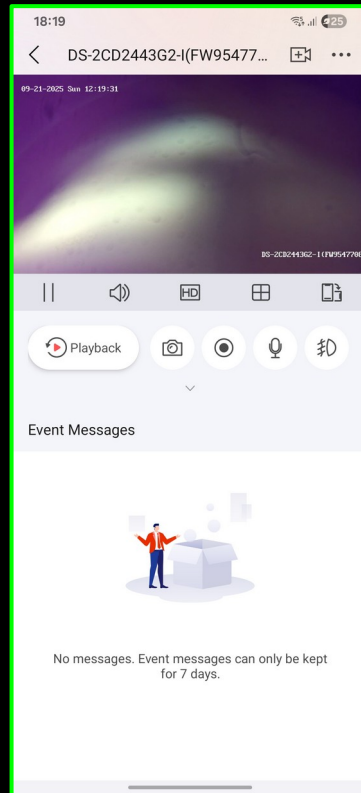
the device

[model] DS-2CD2443G2-I

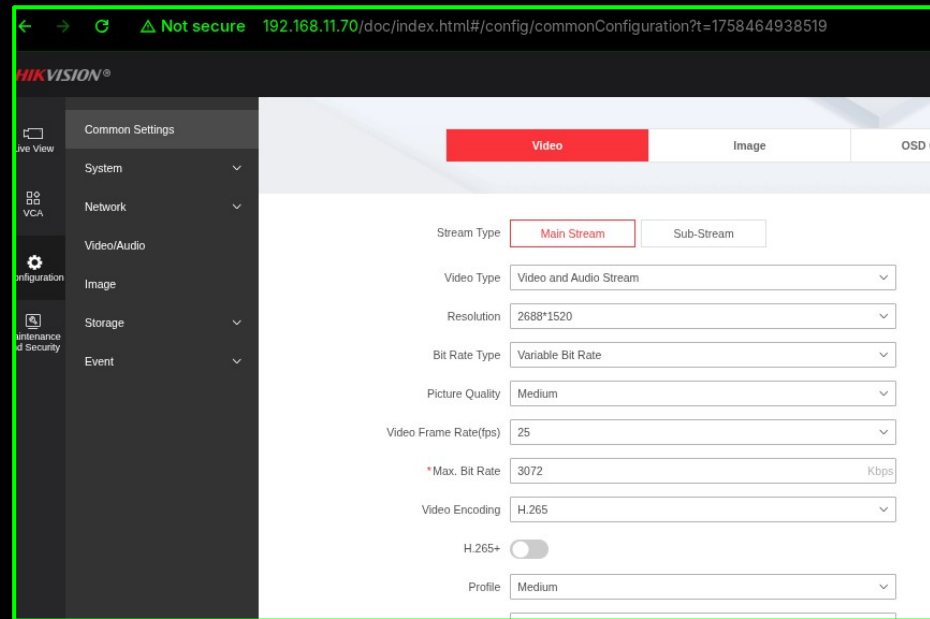
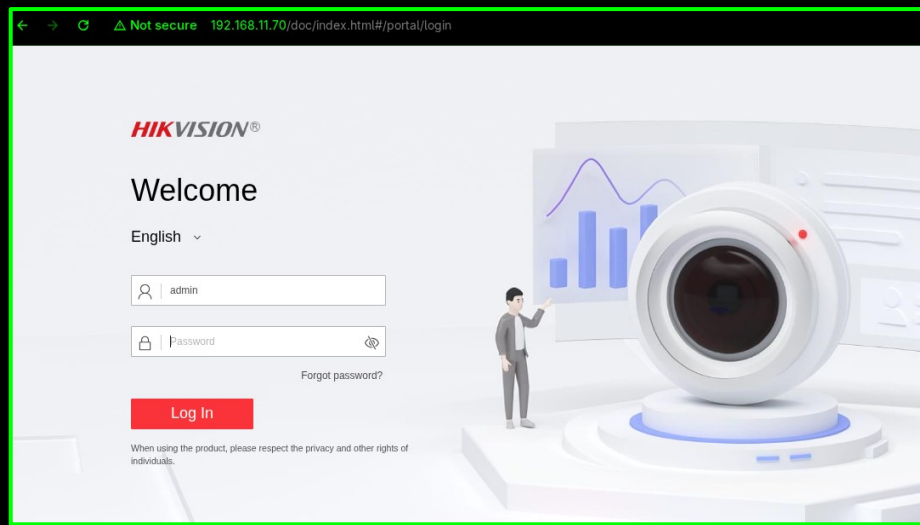
[price] ~80\$ (~100\$)



mobile UI



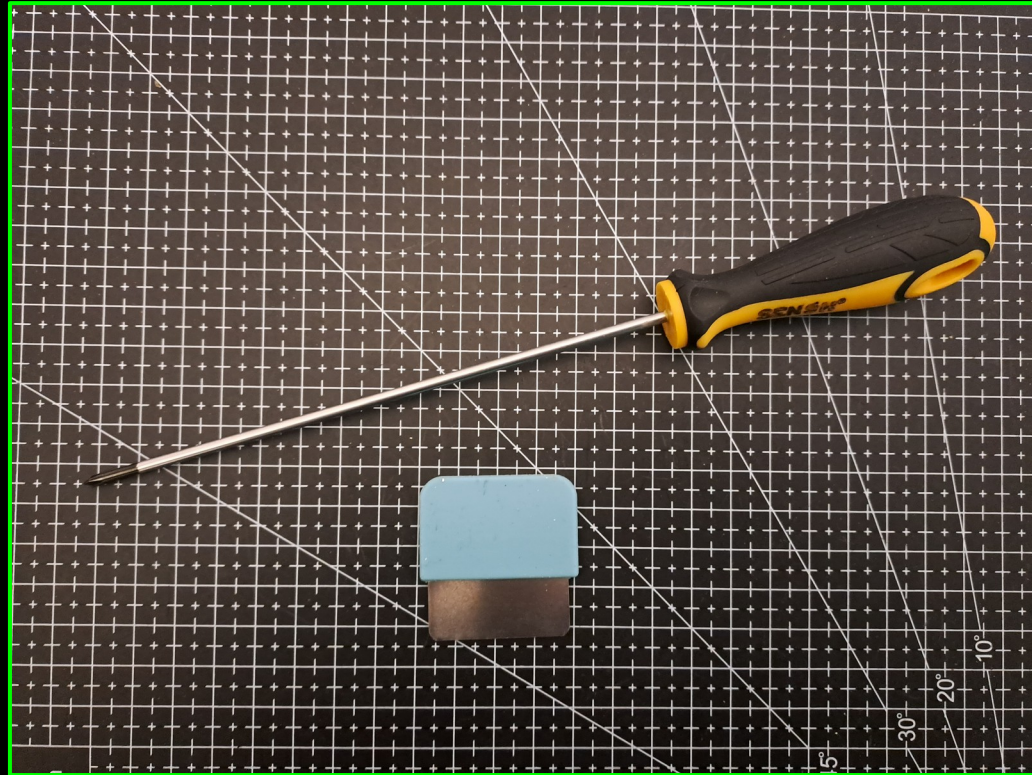
web UI



teardown tools

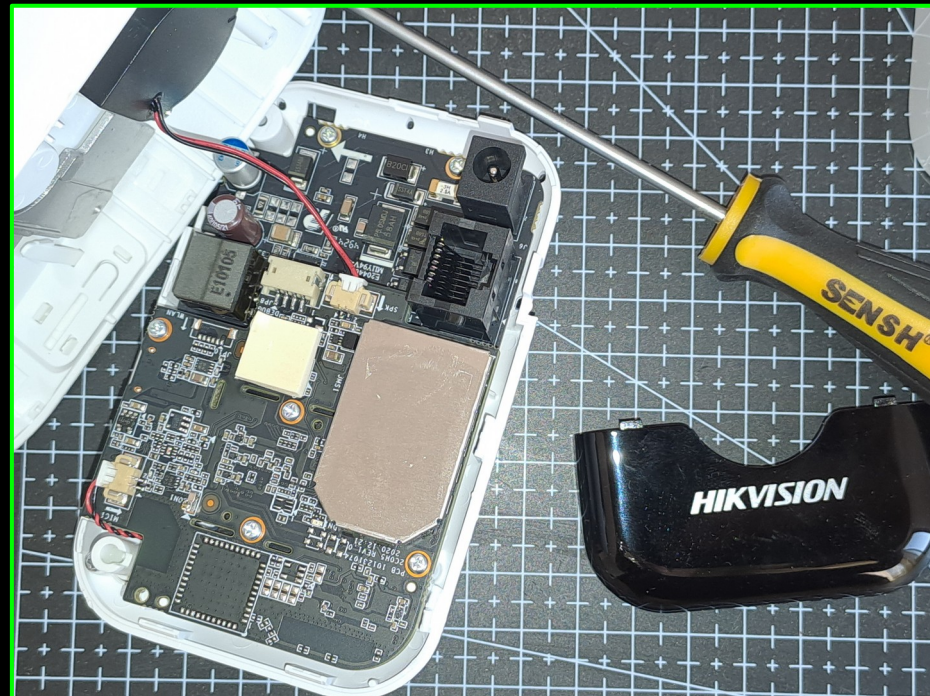
[*] PH Screwdriver

[*] Prying tool



teardown steps

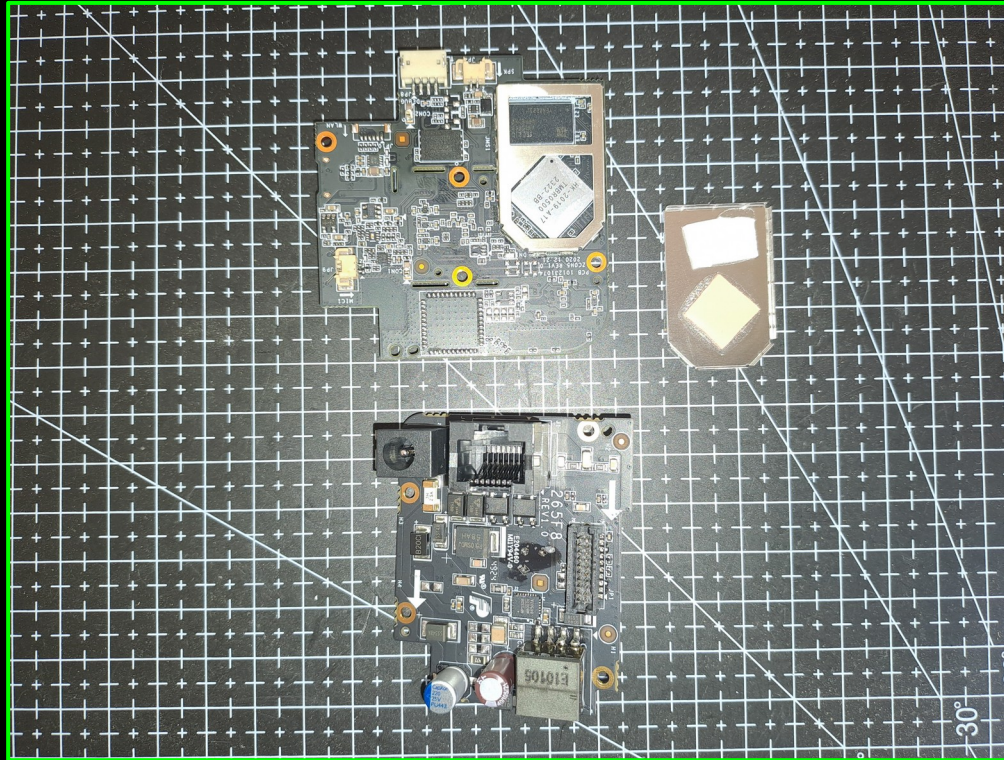
- [*] Remove the bottom housing cover
- [*] Unscrew the rest



PCBS

[*] Logic

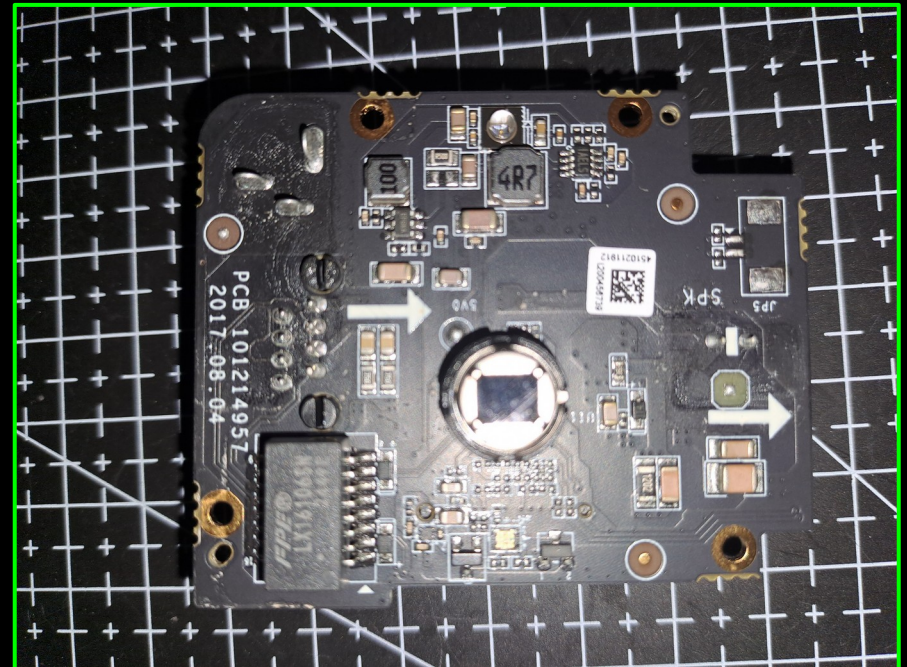
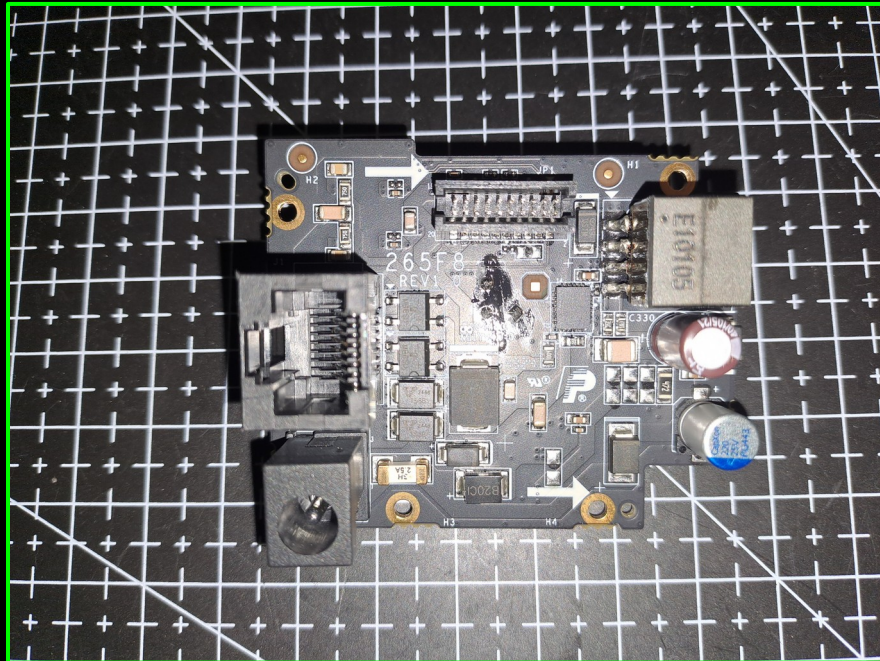
[*] Power



1st PCB

[*] Power connectors

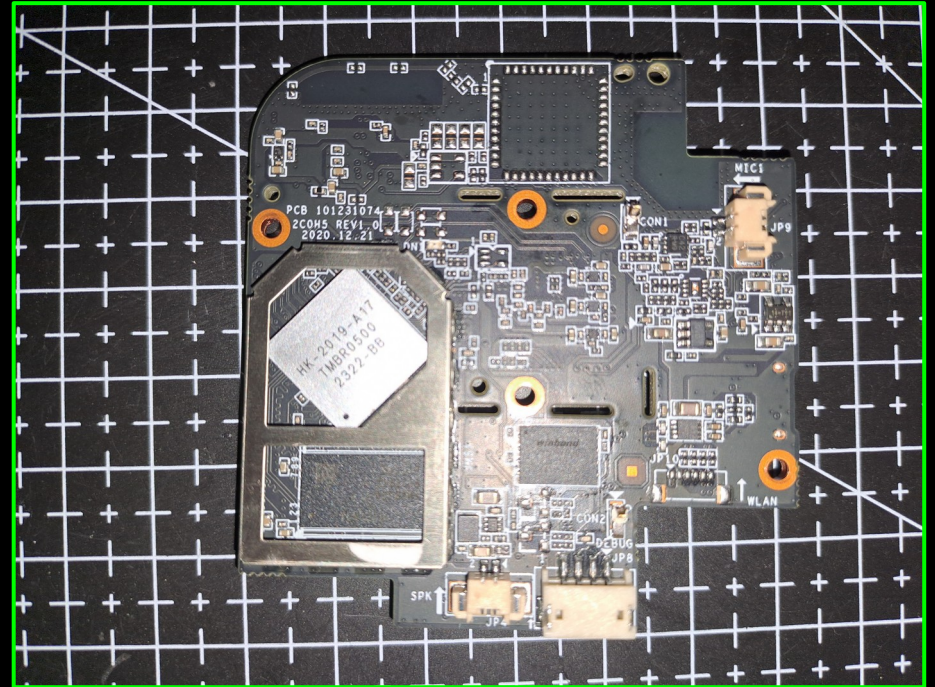
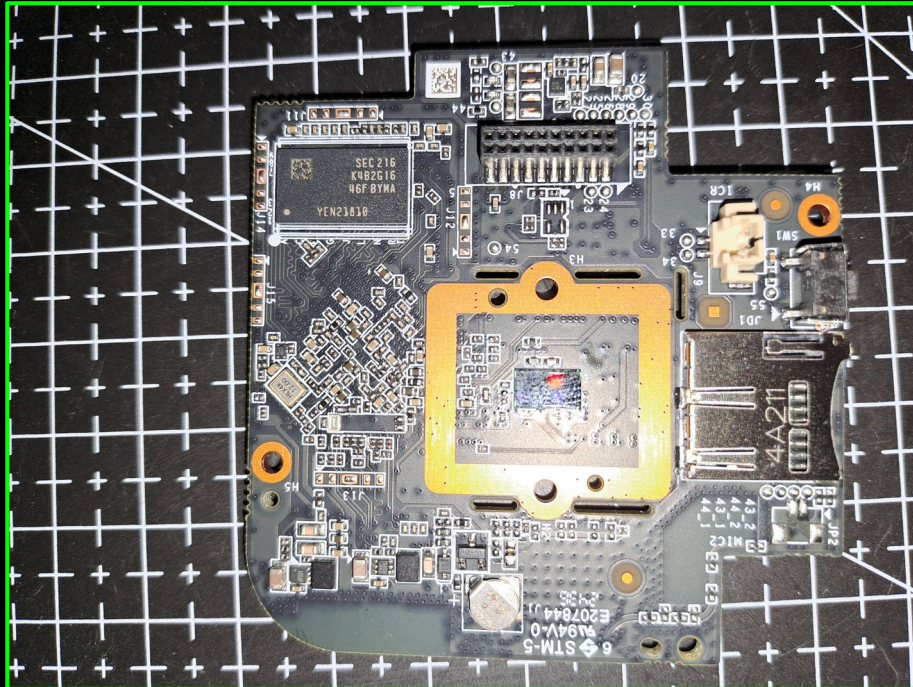
[*] Motion sensor



2nd PCB

[*] Contains the SoC

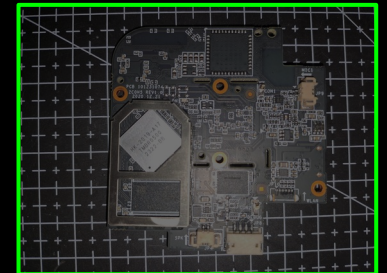
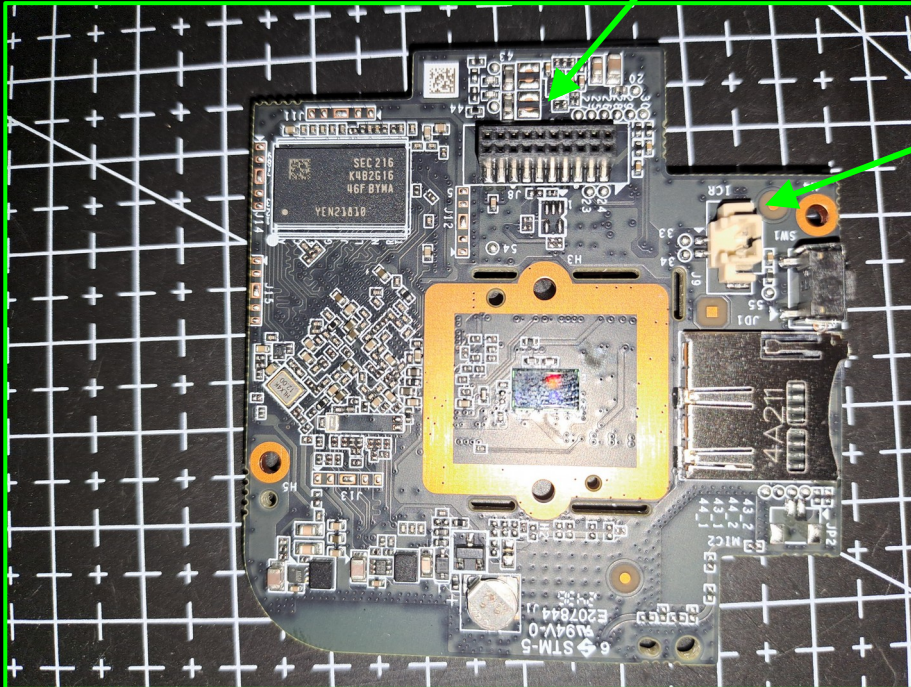
[*] ROM & RAM



connectors

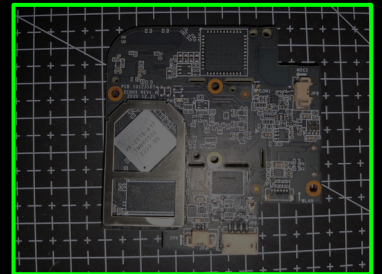
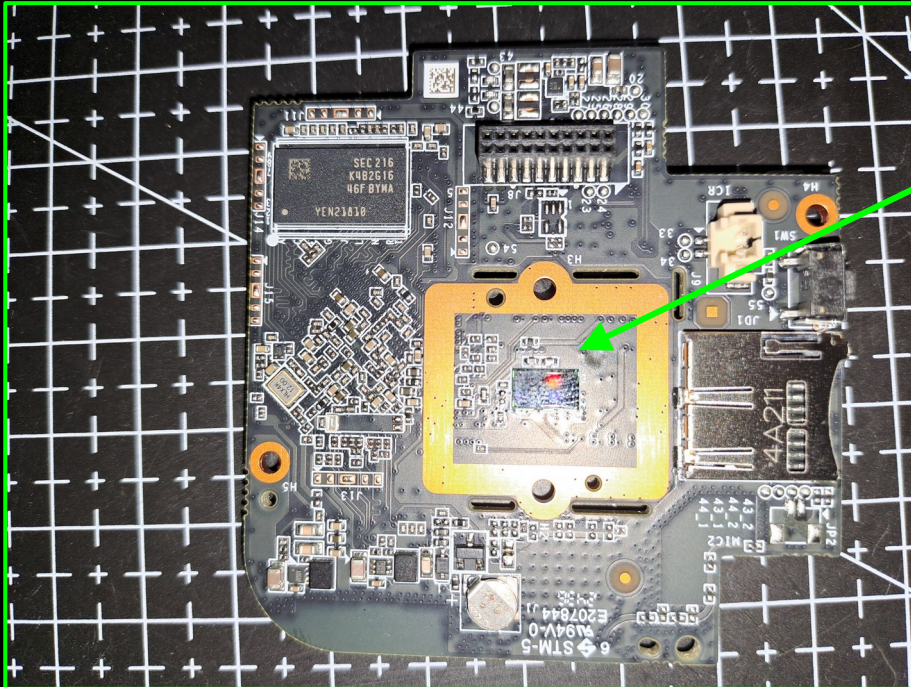
For the 1st board

Camera JST



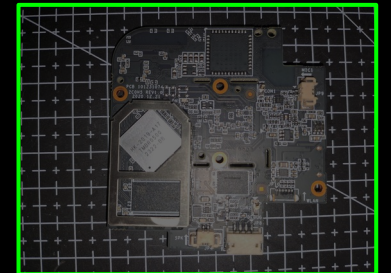
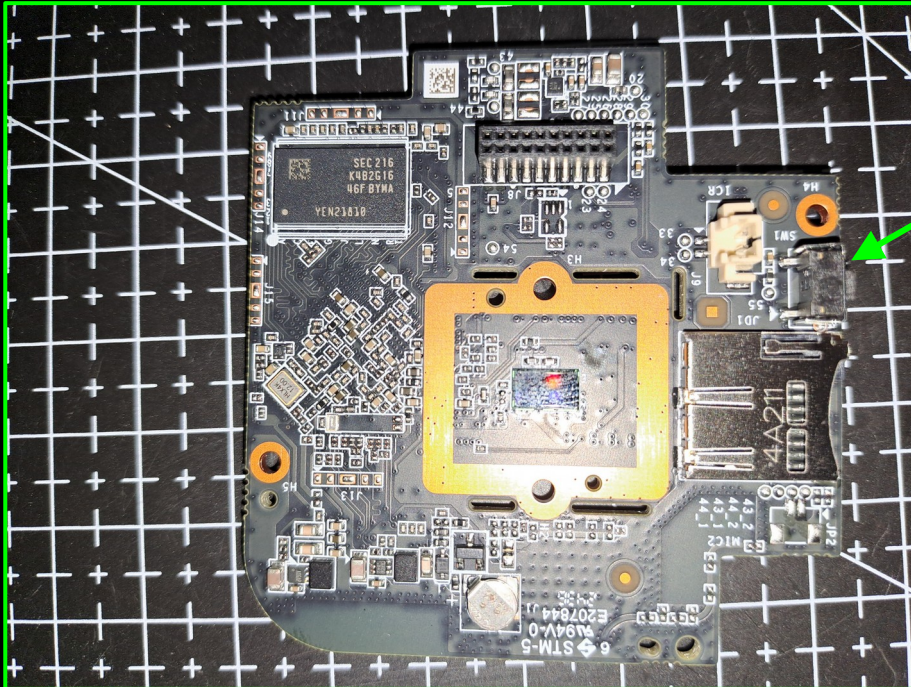
camera

Camera Footprint



buttons

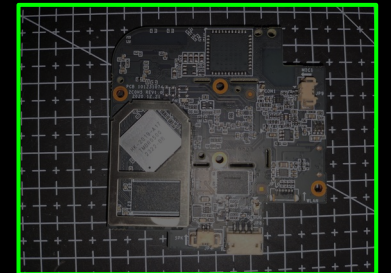
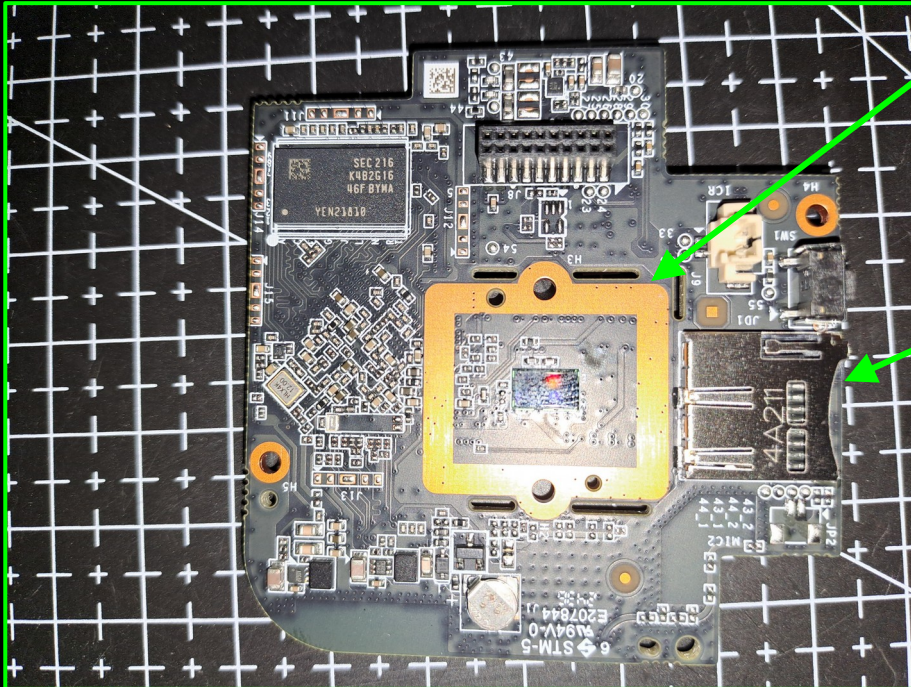
Reset



shielding & ground

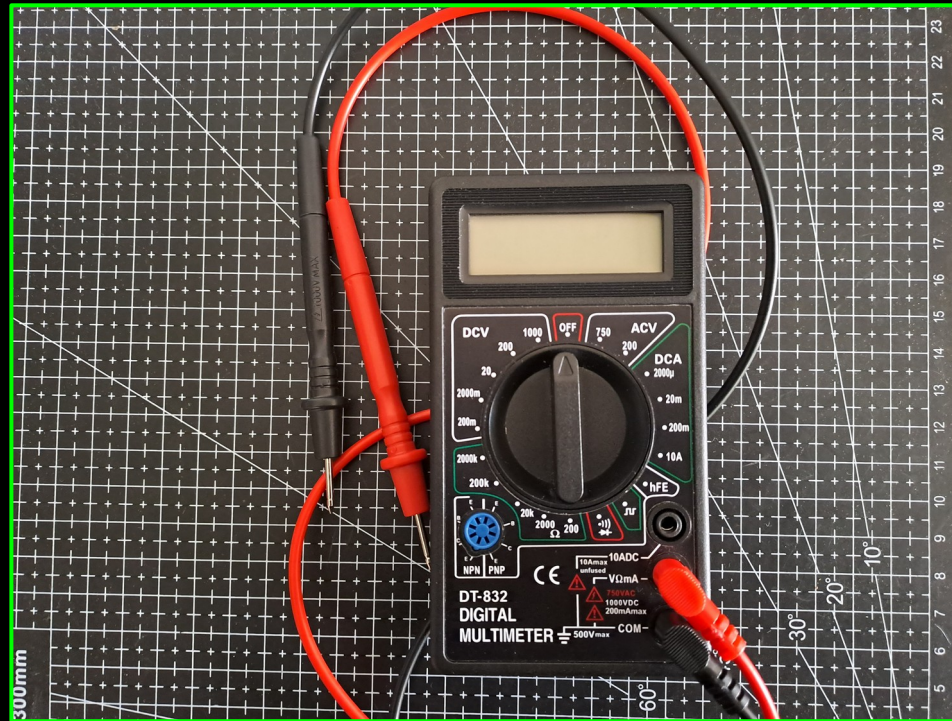
GND

SD Card Slot



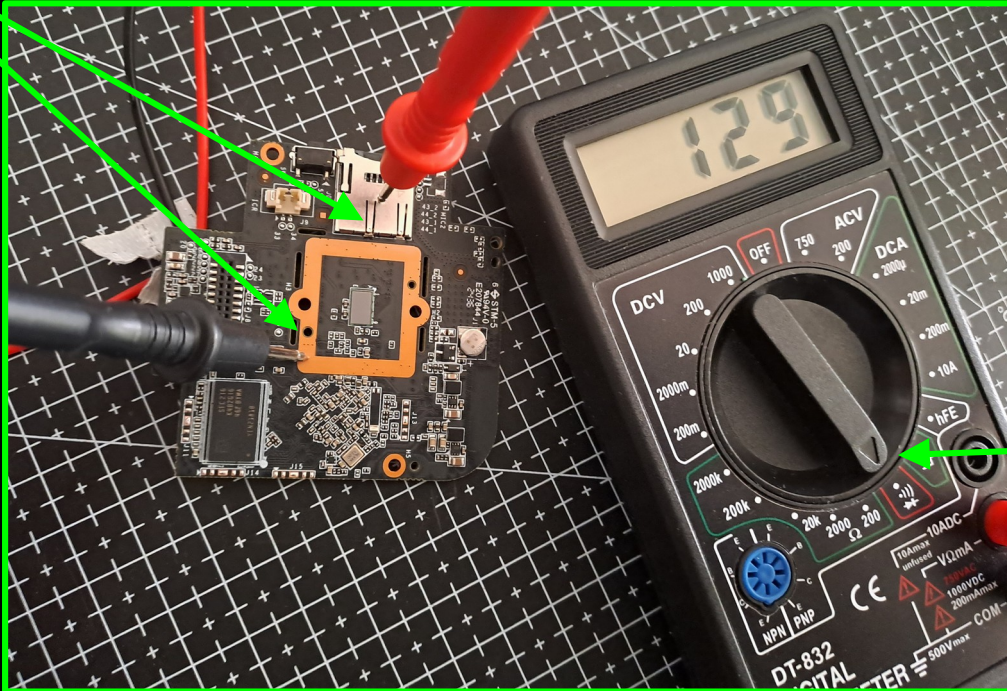
multimeter

- [*] Voltage
- [*] Current
- [*] Resistance
- [*] Continuity
- [*] more...



testing 2 points

probes

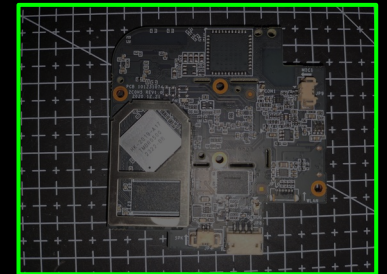
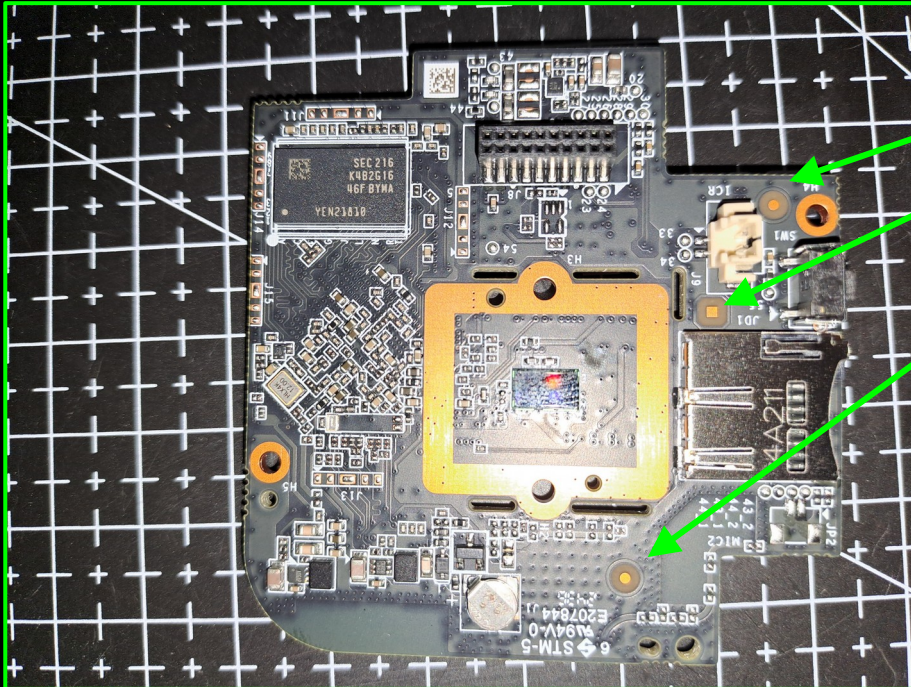


Continuity
mode

pins & pads

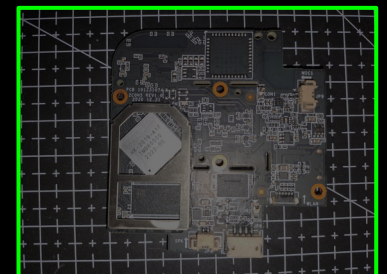
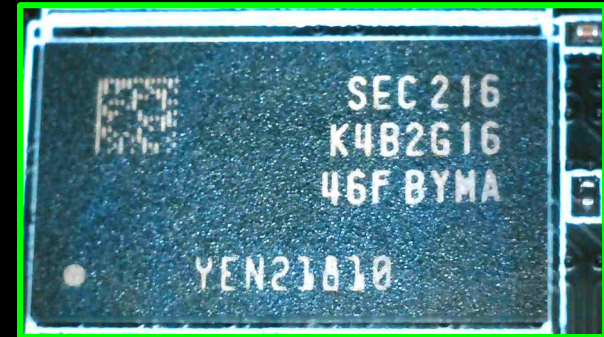
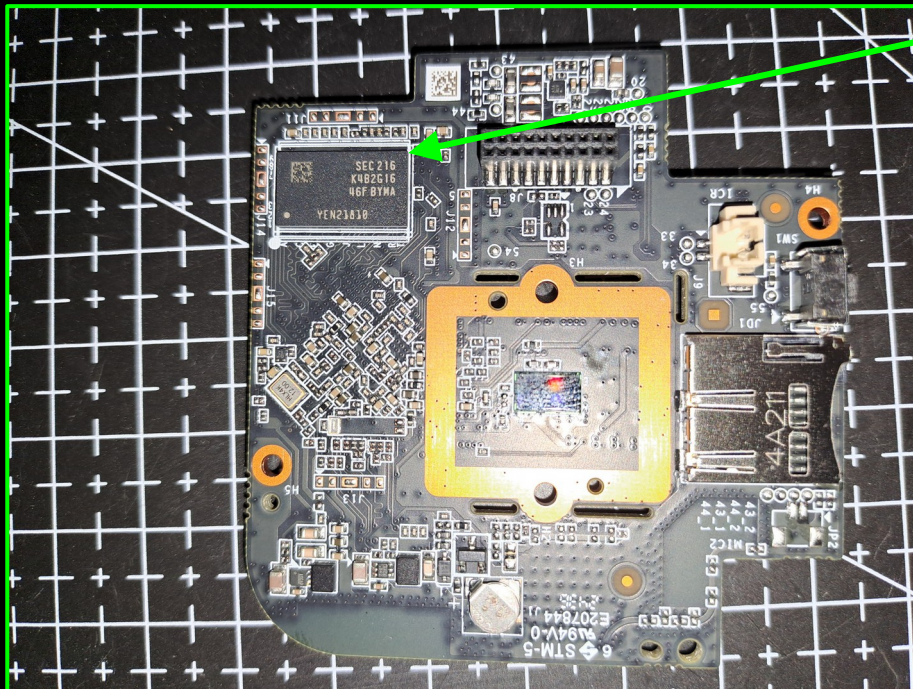
Pads

- [*] Oscilloscope
- [*] Logic Analyzer
- [*] Protocol Analyzer



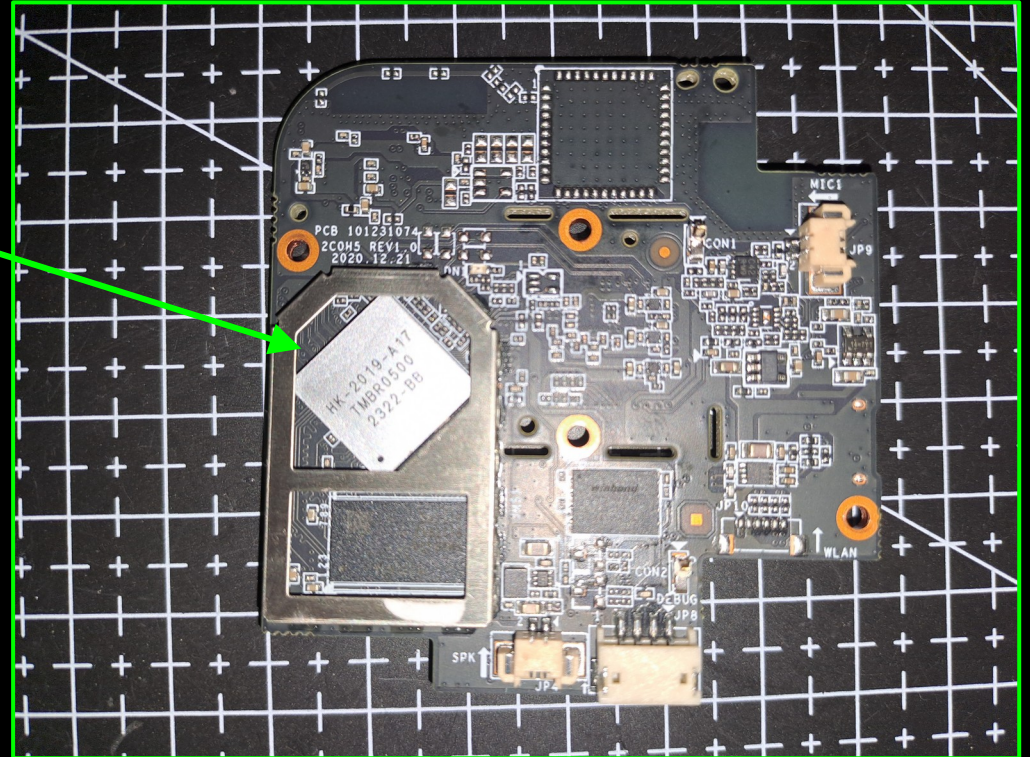
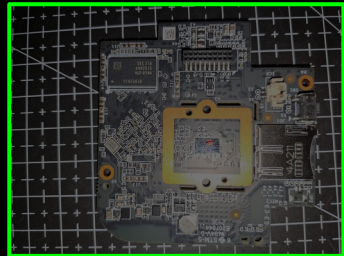
RAM

Samsung SDRAM



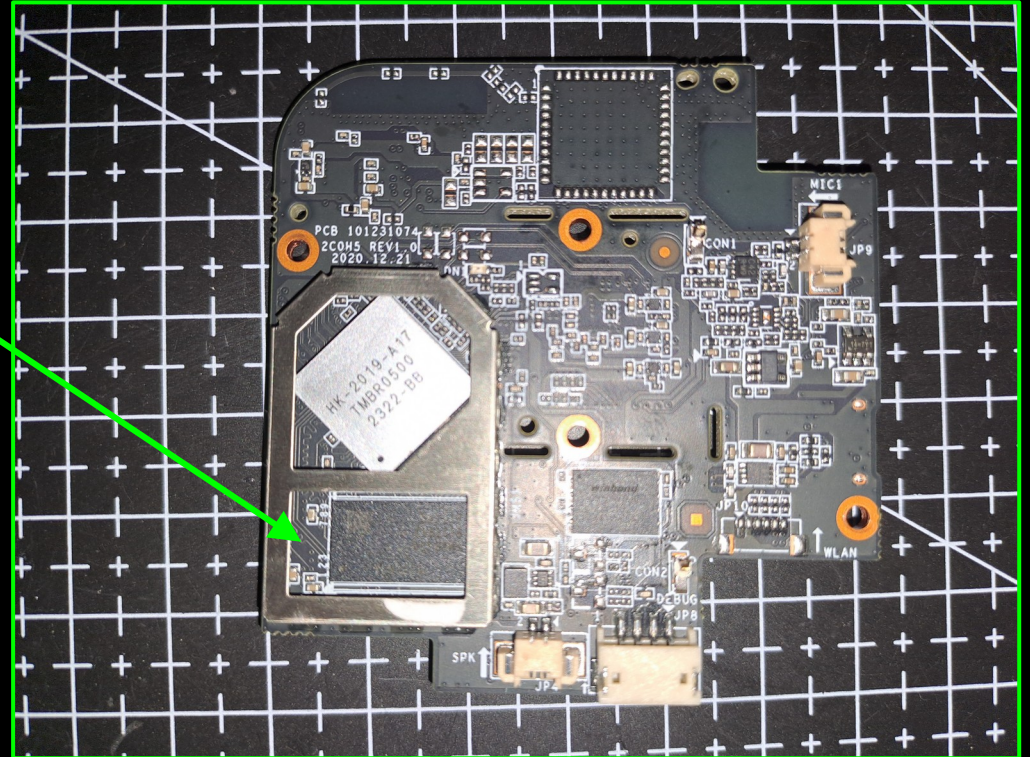
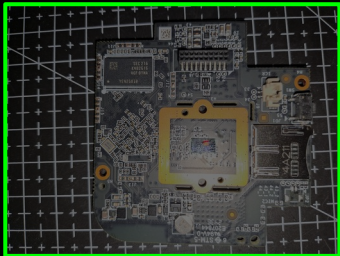
SoC

ARM Cortex



RAM

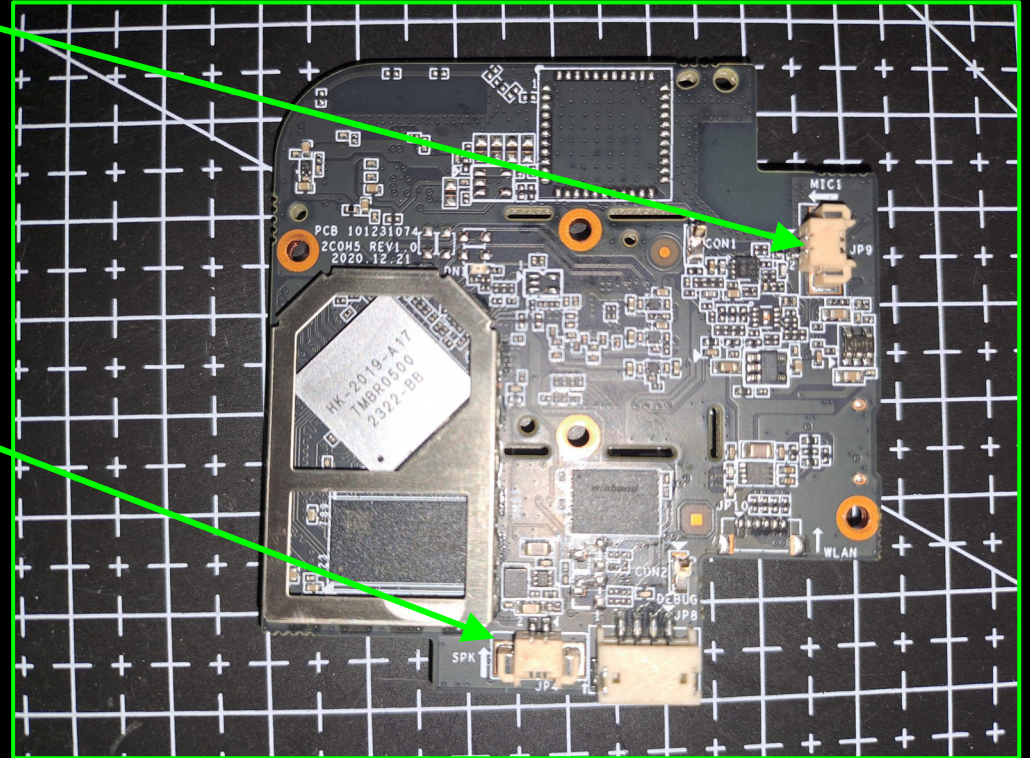
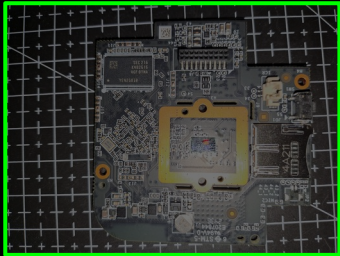
Samsung SDRAM



connectors

Microphone

Speaker

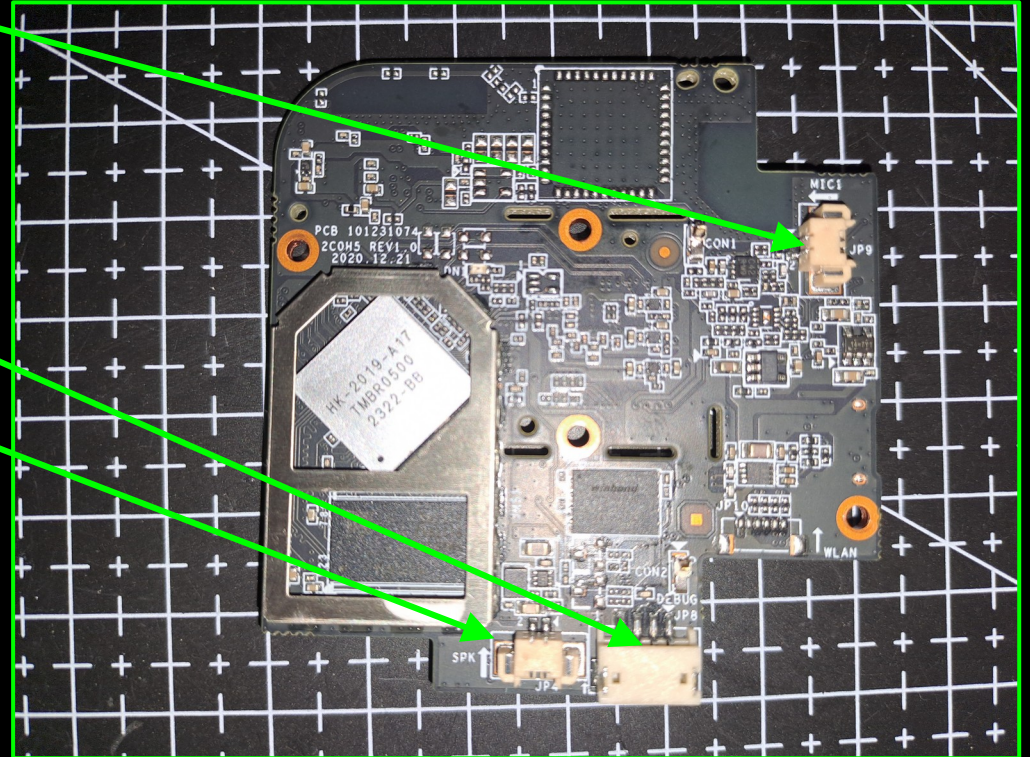
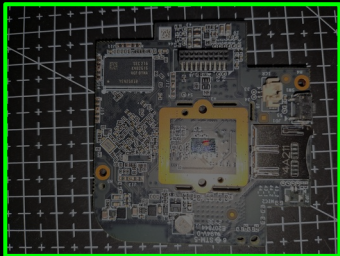


connectors

Microphone

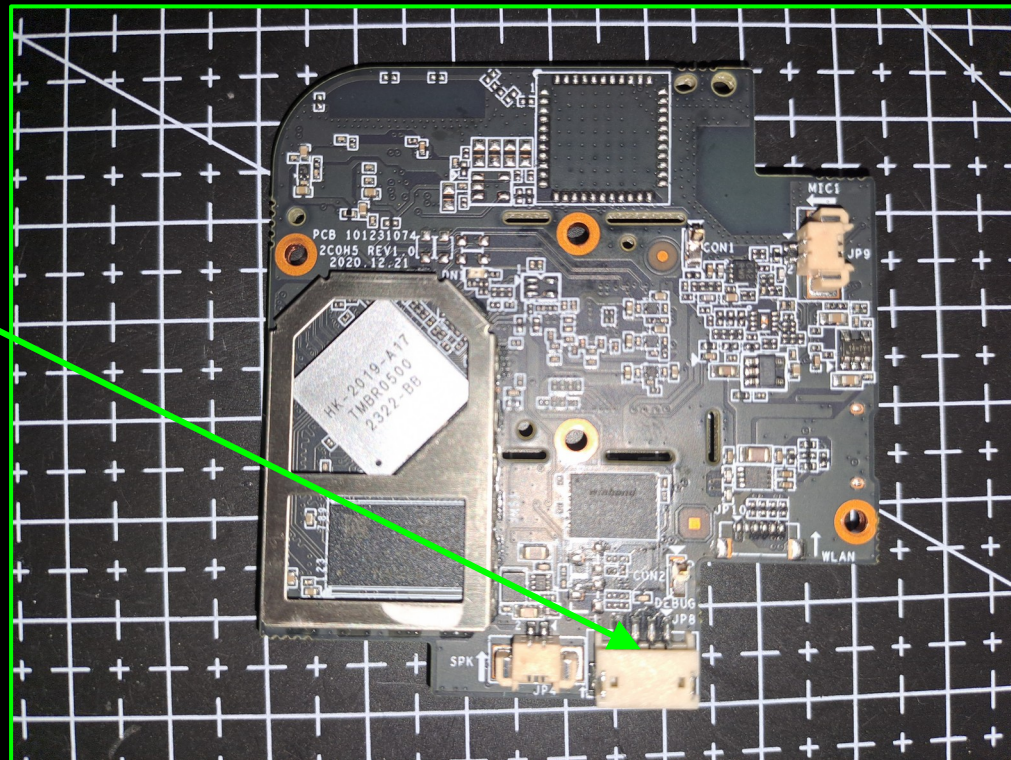
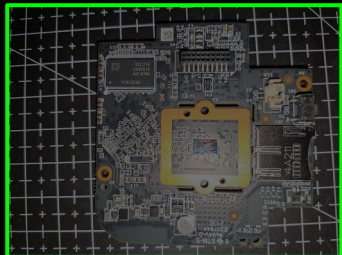
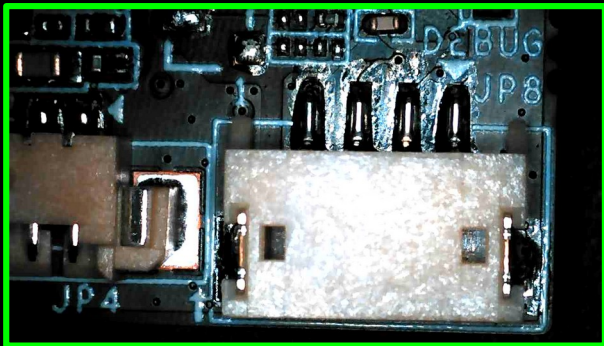
????????

Speaker



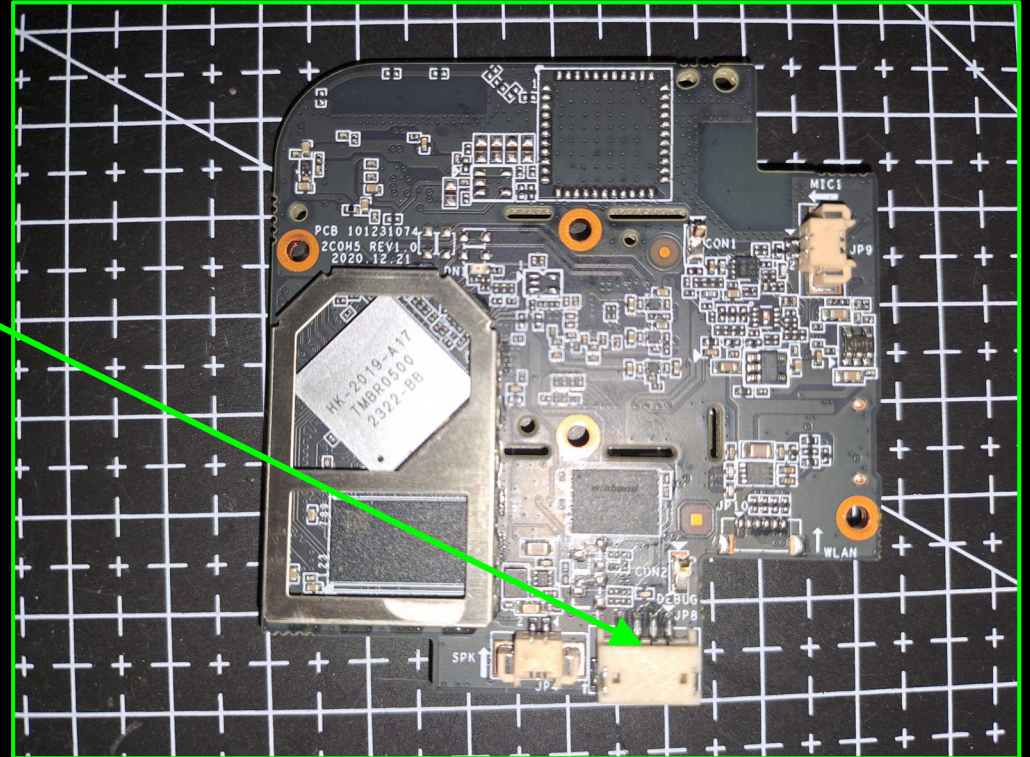
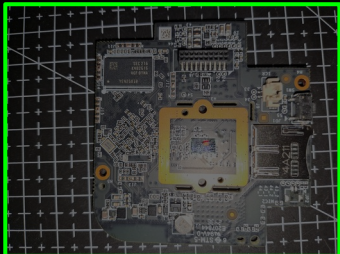
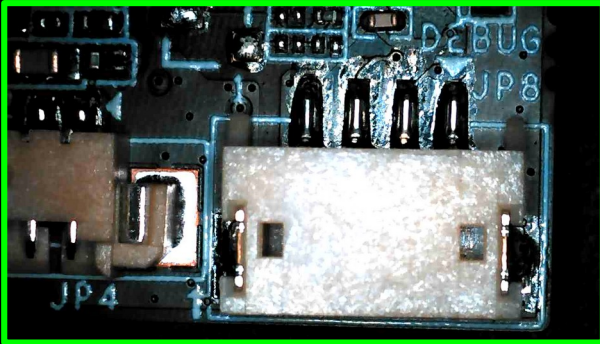
debug

DEBUG? JP8?



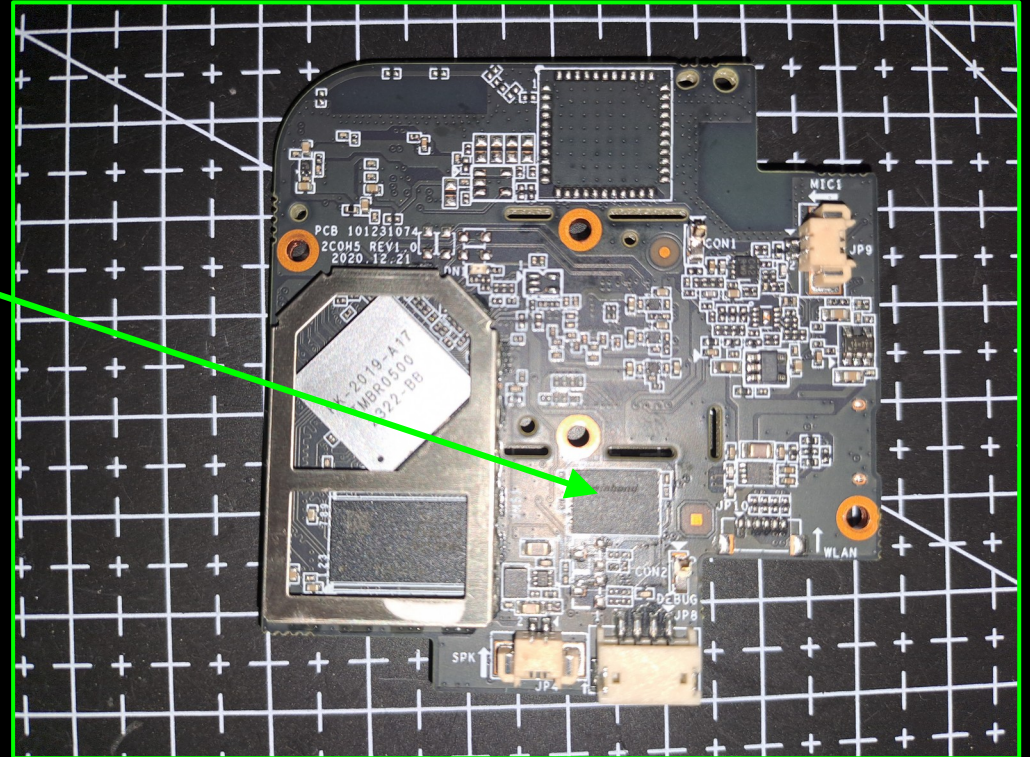
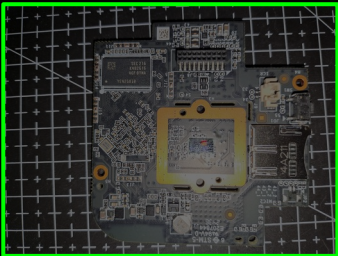
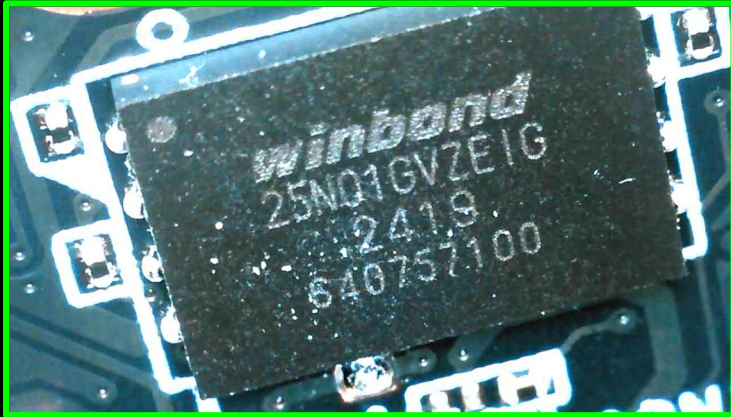
uart

[*] access to the OS



flash

Winbond ROM



summary

- [-] SoC (**S**ystem **o**n a **C**hip)
 - [*] ARMv7 architecture
 - [*] Most likely custom made
- [+] ROM (**R**ead **O**nly **M**emory)
- [+] UART (**U**niversal **A**synchronous **R**eceiver/**T**ransmitter)

summary

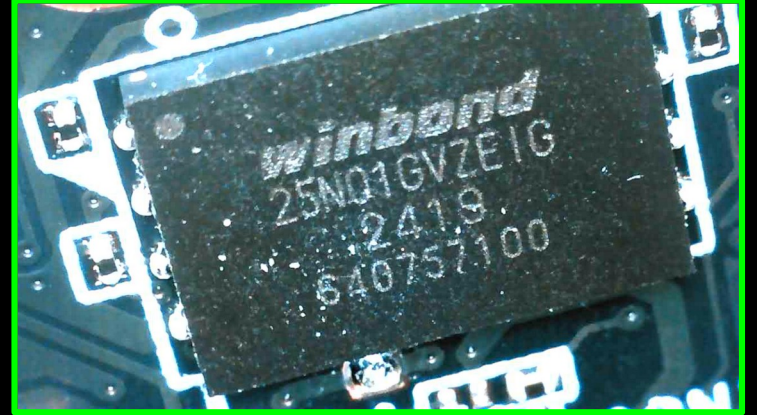
- [+] SoC (**S**ystem **o**n a **C**hip)
- [-] ROM (**R**ead **O**nly **M**emory)
 - [*] Winbond is the manufacturer
 - [*] It can store 128M data
 - [*] Has our 'entire' firmware
 - [*] No security features can be found
- [+] UART (**U**niversal **A**synchronous **R**eceiver/**T**ransmitter)

summary

- [+] SoC (**S**ystem **o**n a **C**hip)
- [+] ROM (**R**ead **O**nly **M**emory)
- [-] UART (**U**niversal **A**synchronous **R**eceiver/**T**ransmitter)
 - [*] Can be used to access the OS shell
 - [*] Usually locked
 - [*] 115200 baudrate (by default)

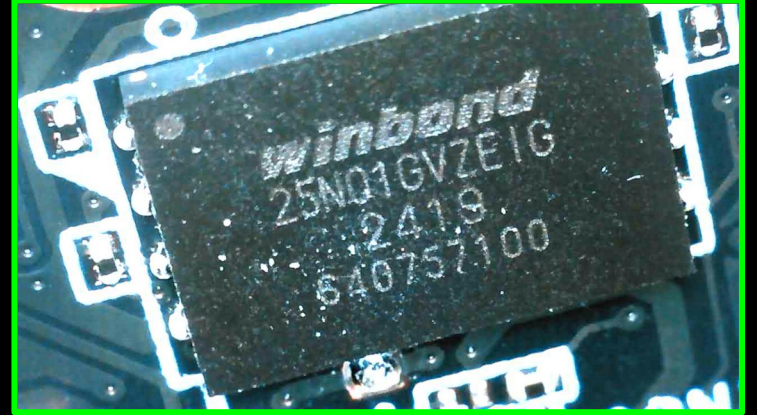
spi flash

[*] 128M storage size
[*] SPI protocol



spi flash

- [*] 128M storage size
- [*] SPI protocol
- [*] WSON8 package



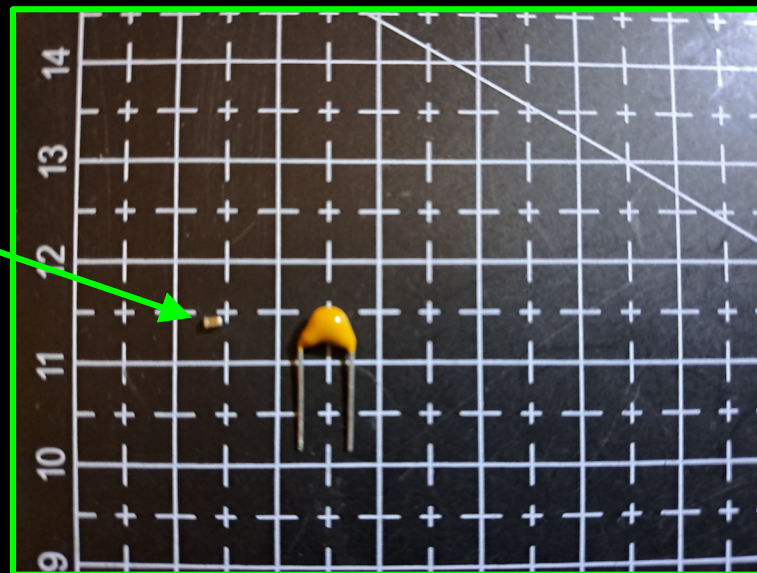
package types

[+] THT (Through-Hole Technology)



package types

- [+] THT (Through-Hole Technology)
- [+] SMT (Surface Mount Technology)

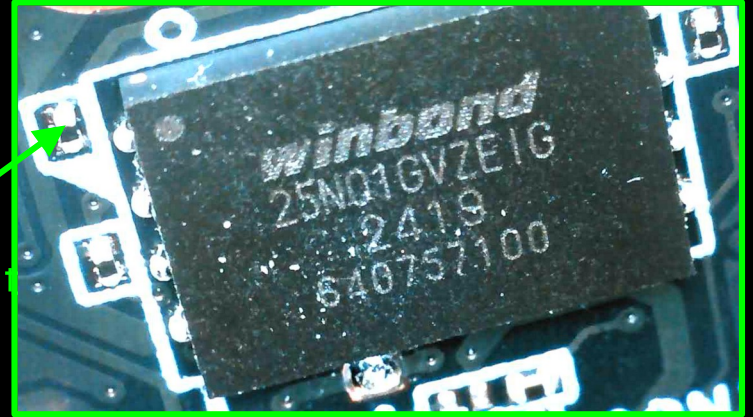


package types

- [+] THT (Through-Hole Technology)
- [-] SMT (Surface Mount Technology)
 - [*] SOP (Small Outline Package)
 - [*] QFN (Quad Flat No-lead)
 - [*] BGA (Ball Grid Array)
 - [*] WSON (Wettable-flank Small Outline No-lead)
 - [*] ...

spi flash

- [*] 128M storage size
- [*] SPI protocol
- [*] WSON8 package
- [*] Small components can be found around it



desoldering

- [*] provides true isolation
- [*] its very tedious
- [*] special packages (e.g.: BGA) are not easy to solder



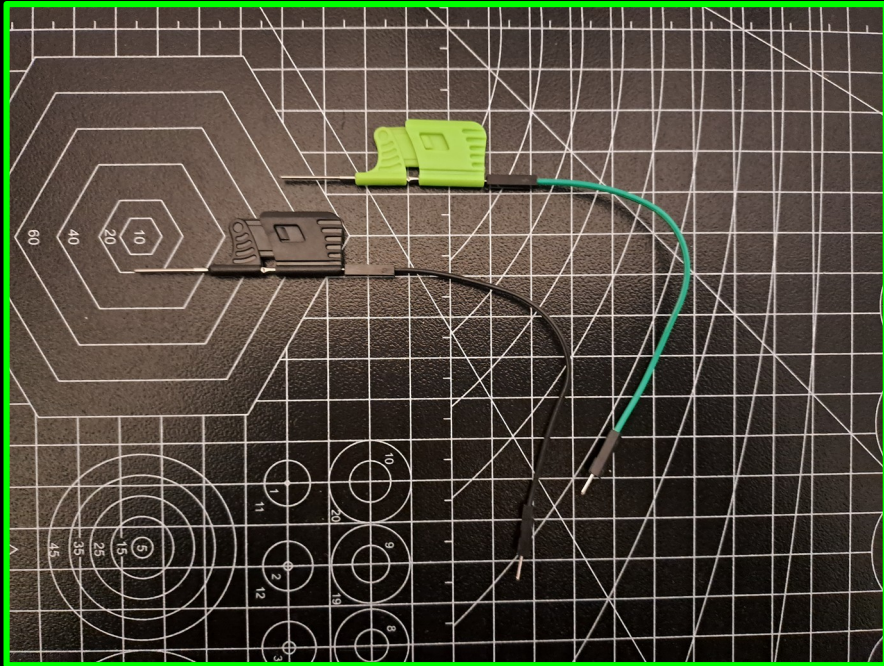
desoldering

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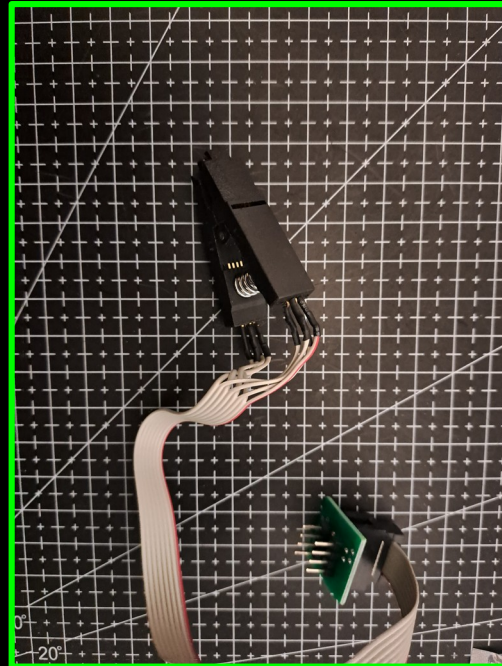
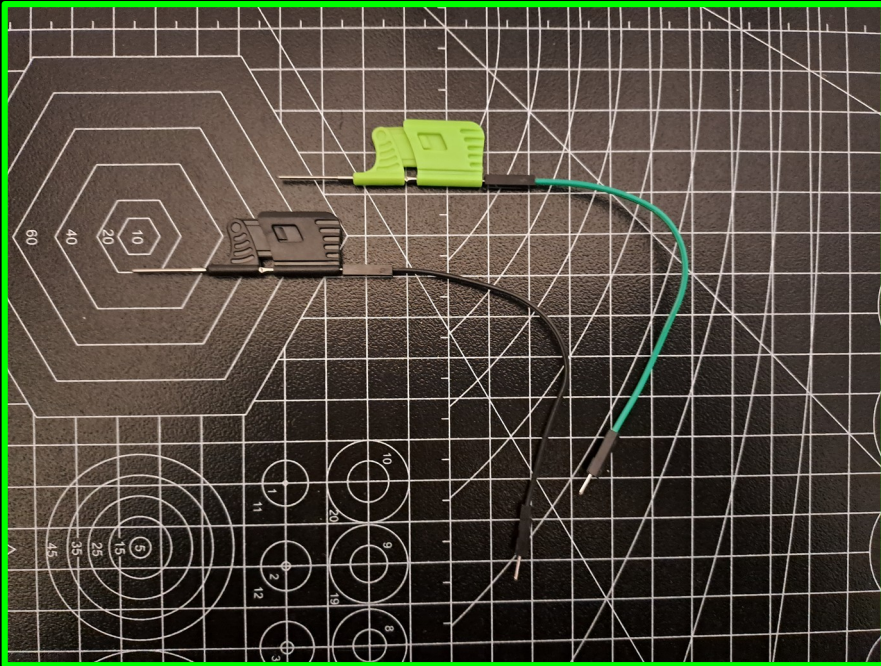
probing

- [*] quick and easy
- [*] does not work with BGA packages
- [*] no protection for reverse current



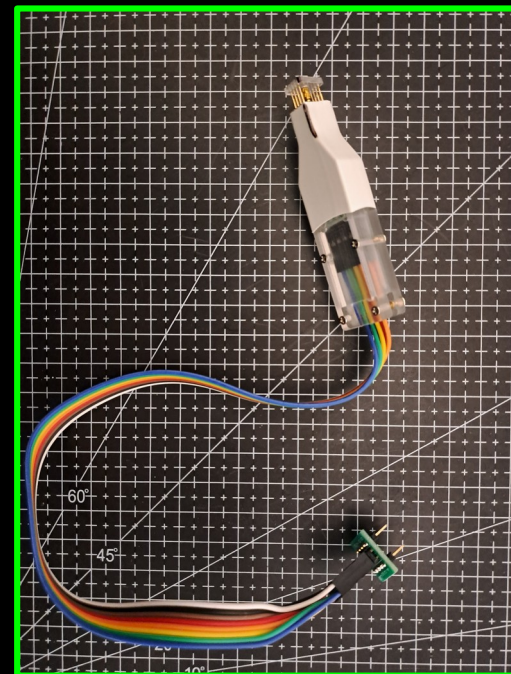
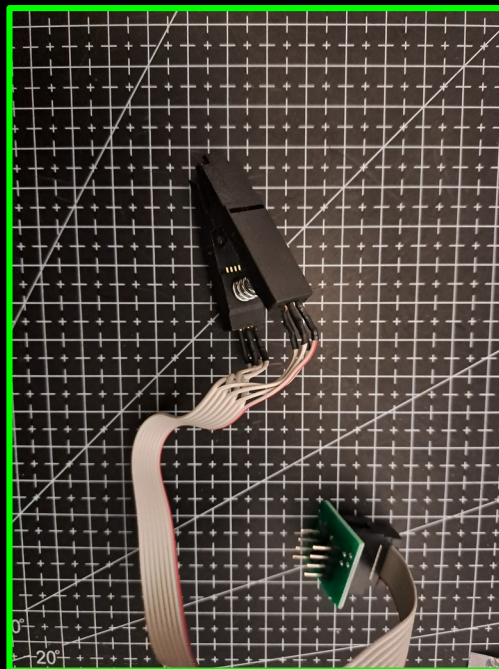
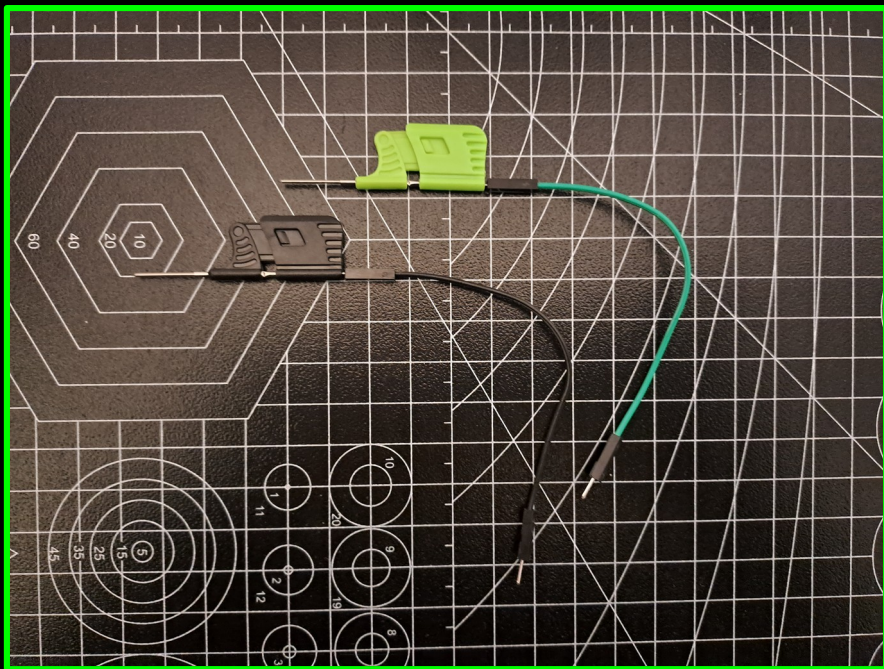
probing

- [*] quick and easy
- [*] does not work with BGA packages
- [*] no protection for reverse current



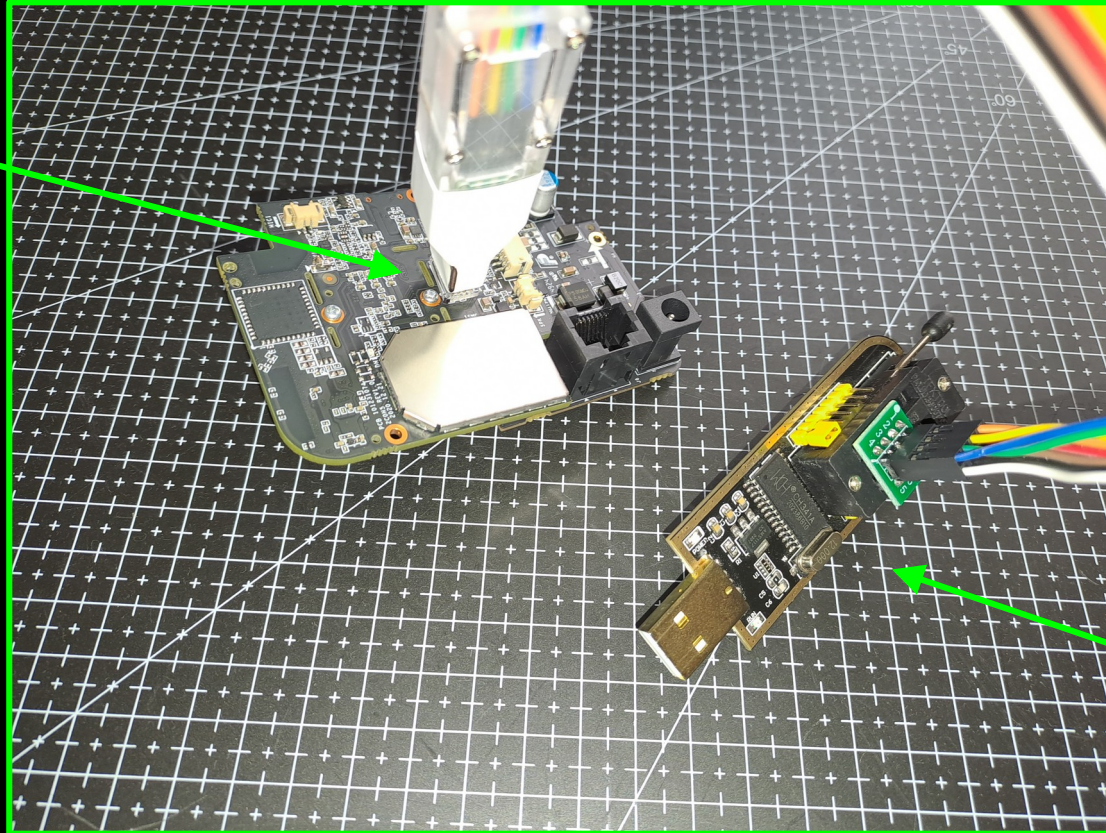
probing

- ```
[*] quick and easy
[*] does not work with BGA packages
[*] no protection for reverse current
```



# sbi setup

SPI Flash



CH341A

# spi reader software

|            |                             |
|------------|-----------------------------|
| [flashrom] | will address 99% of cases   |
| [snander]  | will cover 0.99% of cases   |
| [custom]   | for 0.01%, use your own MCU |

# snander

```
$ snander -r firmware
```



# snander

```
$ snander -r firmware
$ snander -r firmware.2
```

# snander

```
$ snander -r firmware
```

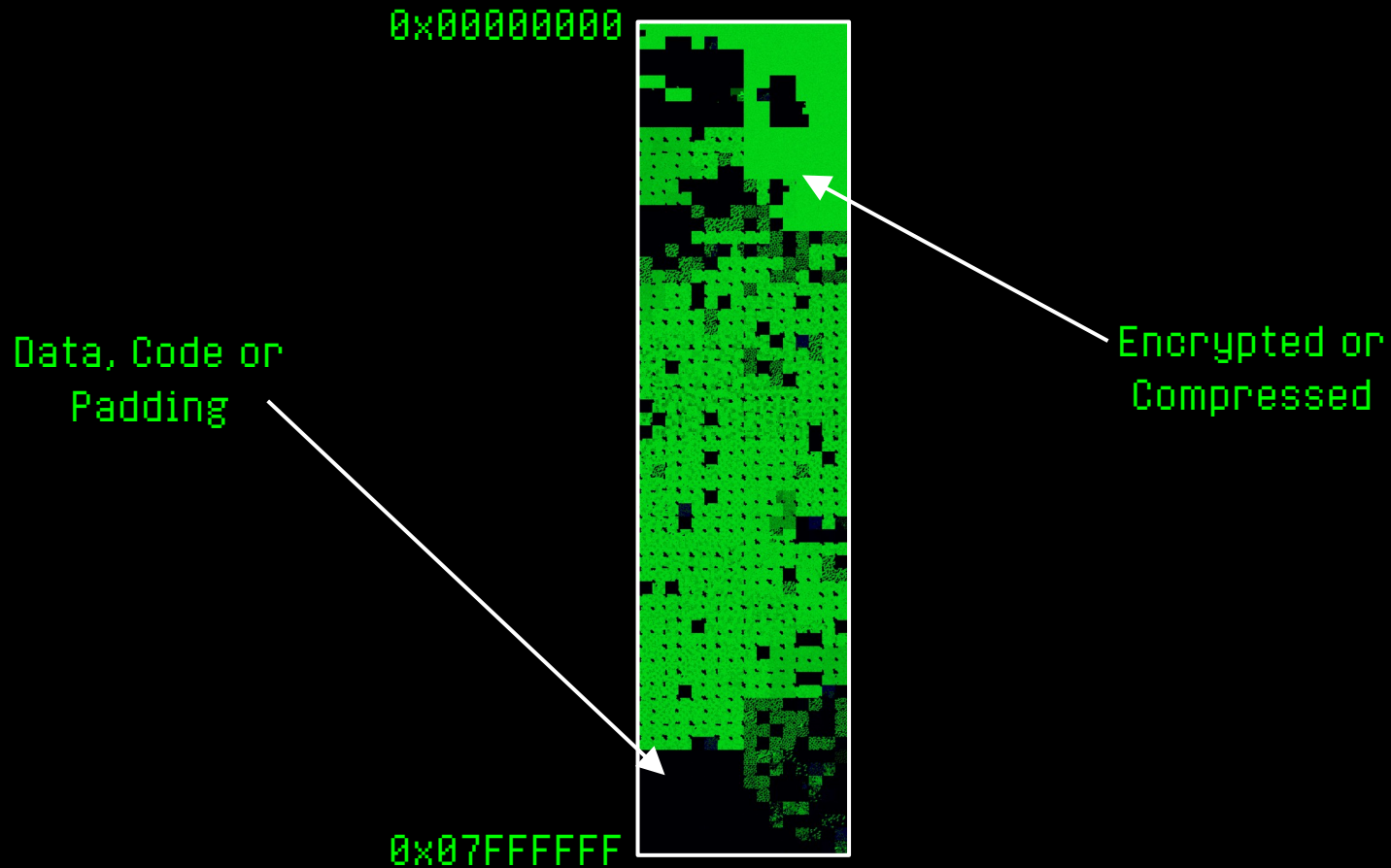
```
$ snander -r firmware.2
```

```
$ md5sum firmware*
```

```
b2048ed3f684ae33458e440a86252774 firmware
```

```
b2048ed3f684ae33458e440a86252774 firmware.2
```

# entropy



# binwalk

```
$ binwalk -Me firmware
```

```

HEXADECIMAL DESCRIPTION

0x07000000 Device tree blob (DTB)
0x0F000000 Device tree blob (DTB)
0x17000000 UBI image, version: 1
0x6B000000 UBI image, version: 1
0x73400000 UBI image, version: 1
0x76000000 UBI image, version: 1
...
0x78C00000 UBI image, version: 1
0x79000000 UBI image, version: 1

```



# device tree

```
$ grep "nand" system.dts
```

```
nand@f0400000 {
 ...
 partition_bld {
 label = "bld";
 reg = <0x00 0x00 0x00 0x200000>;
 };
 partition_teeos0 {
 label = "teeos0";
 reg = <0x00 0x200000 0x00 0x180000>;
 };
};
```

# firmware structure



# enviornment partition

```
$ dd if=firmware skip=$((0x500000)) bs=1 | hexdump -C | head
00000000 52 e2 b8 71 61 72 63 68 3d 61 72 6d 00 62 61 75 IR..qarch=arm.baul
00000010 64 72 61 74 65 3d 31 31 35 32 30 30 00 62 6f 61 ldrate=115200.boal
00000020 72 64 3d 6e 76 74 2d 6e 61 35 31 30 35 35 00 62 lrd=nvt-na51055.bl
00000030 6f 61 72 64 5f 6e 61 6d 65 3d 6e 76 74 2d 6e 61 loard_name=nvt-na1
00000040 35 31 30 35 35 00 62 6f 6f 74 61 72 67 73 3d 65 l51055.bootargs=el
00000050 61 72 6c 79 70 72 69 6e 74 6b 20 63 6f 6e 73 6f larlyprintk consol
00000060 6c 65 3d 74 74 79 53 30 2c 31 31 35 32 30 30 20 lle=ttyS0,115200 l
00000070 72 6f 6f 74 77 61 69 74 20 6e 70 72 6f 66 69 6c lrootwait nprofill
00000080 65 5f 69 72 71 5f 64 75 72 61 74 69 6f 6e 3d 6f le_irq_duration=ol
00000090 6e 20 72 6f 6f 74 3d 75 62 69 30 3a 72 6f 6f 74 ln root=ubi0:rootl
```

# enviornment partition

```
qarch=arm
board=nvt-na51055
board_name=nvt-na51055
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
 root=ubi0:rootfs rootfstype=ubifs ubi
 fm_autoconvert=1 init=/linuxrc boot_mode=0
bootcmd=loadk;bootm
bootdelay=3
cpu=armv7
dbg=0
ethact=eth_hikethaddr=08:cc:81:61:fd:15
fdtcontroladdr=6f9c600
gatewayip=192.168.1.254
hostname=soclnx
ipaddr=192.168.1.71
netmask=255.255.255.0
serverip=192.168.1.128
verify=0
```



# arm trusted firmware

Reset

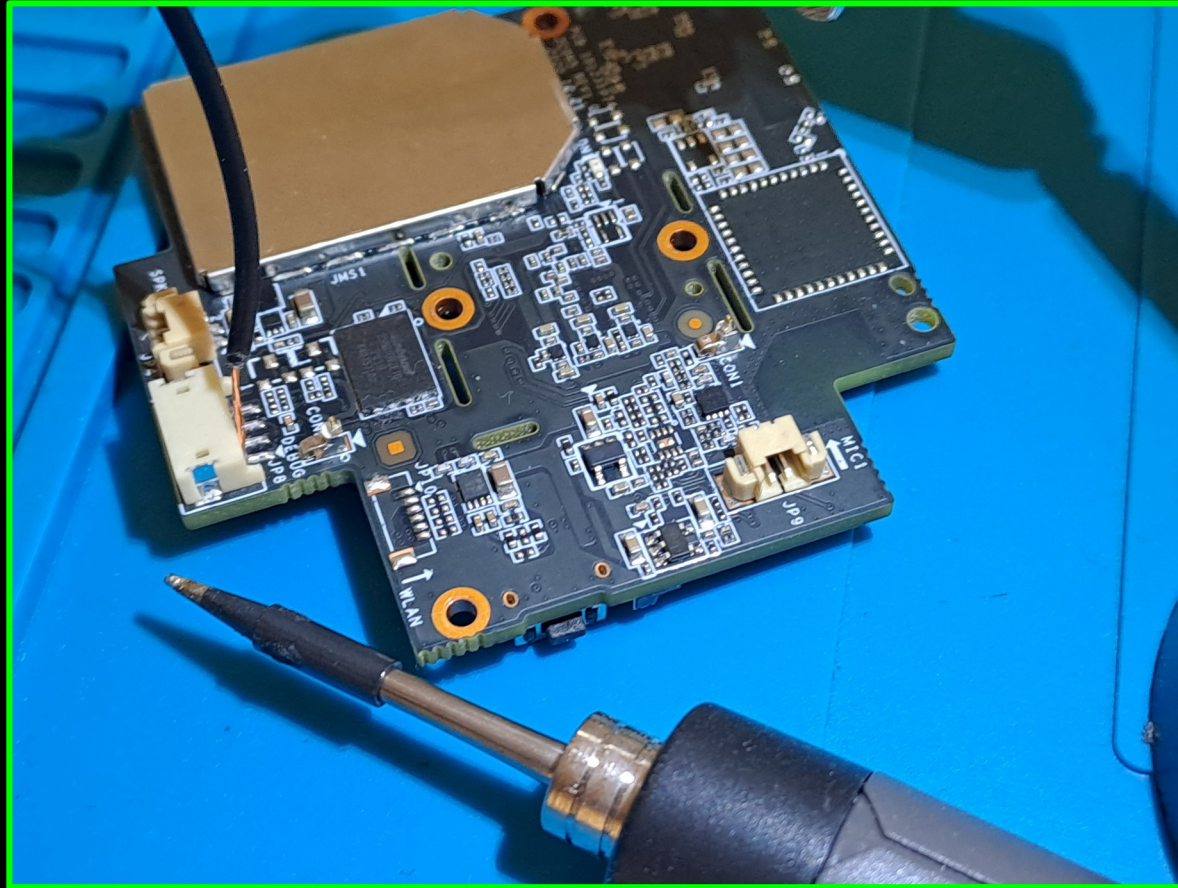
Boot Code [FUSE]

U-Boot/TEEOS [ROM]

TEEOS/U-Boot [ROM]

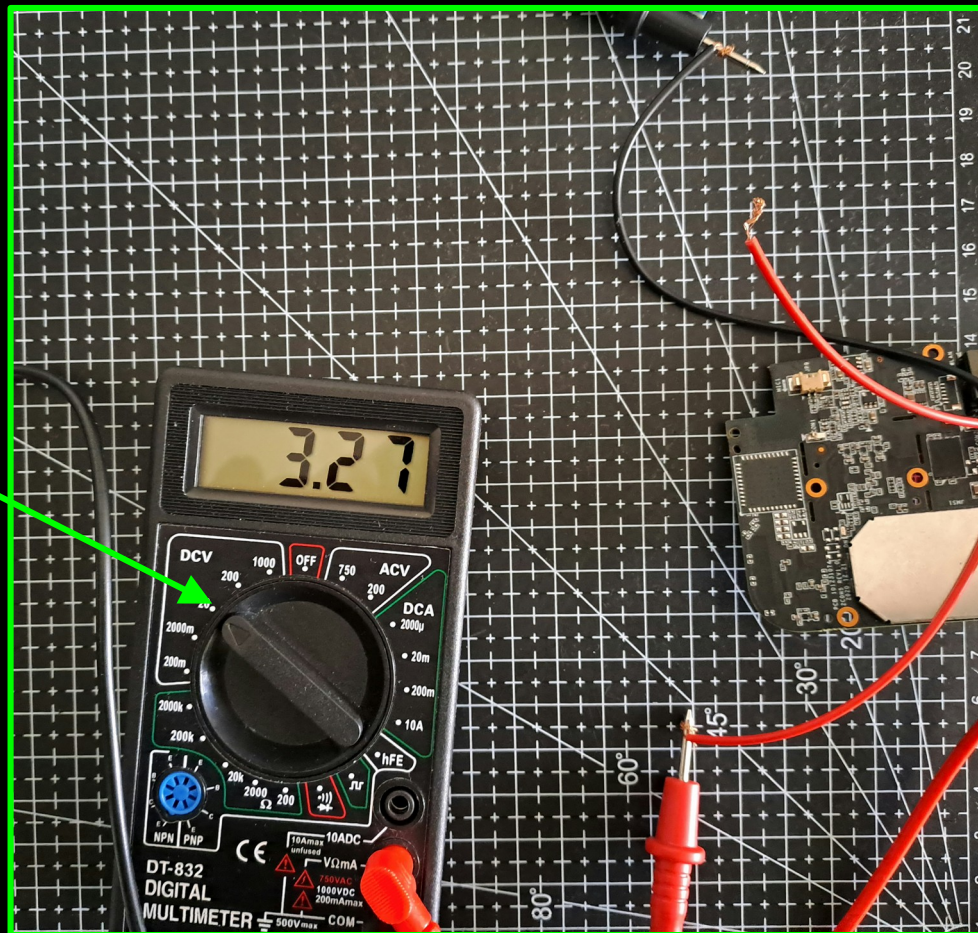
Linux [ROM]

# soldering wires



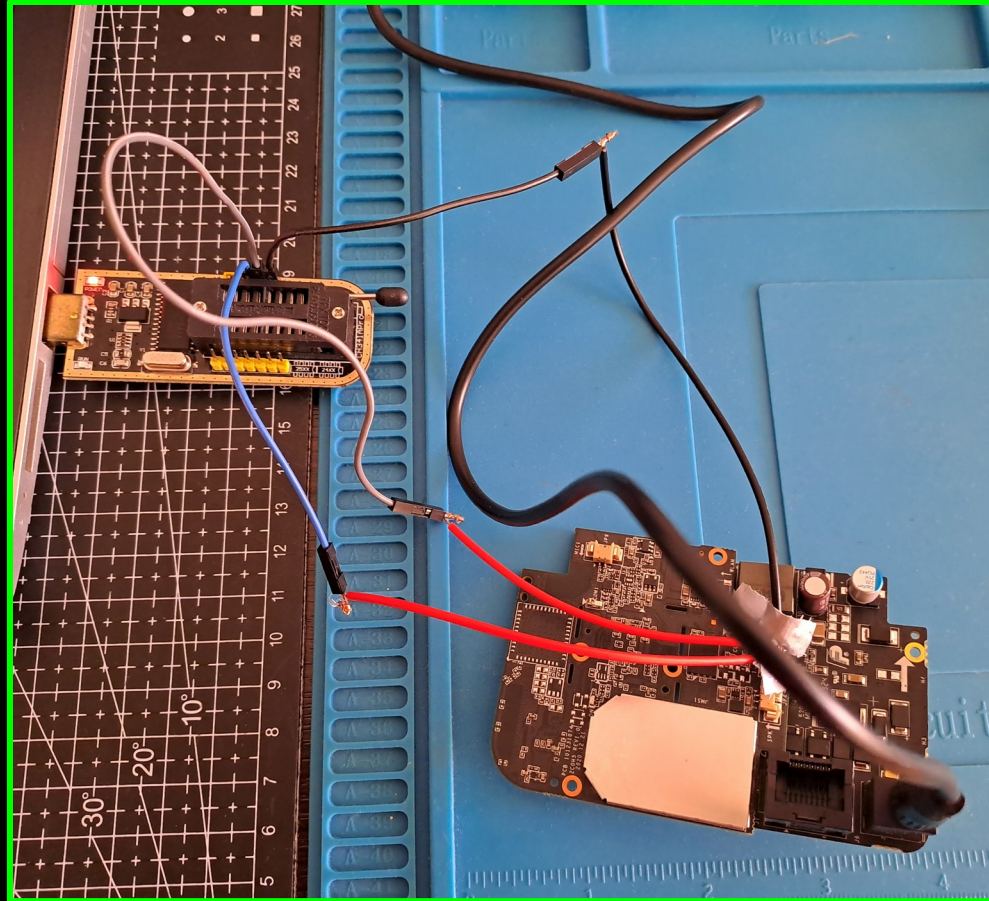
# measuring voltage

Voltage Mode





# uart setup





# serial console

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
[Uboot] In release mode!
```

```
Hit Ctrl+u to stop autoboot: 0
```

```
eth_hik Waiting for PHY auto negotiation to complete..... TIMEOUT !
```

```
...
```

```
TFTP server auto connect disabled
```

```
booting from sys part...
```

```
#
```

# u-boot shell

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
HKUS# printenv
```

```
arch=arm
```

```
baudrate=115200
```

```
board=nvt-na51055
```

```
board_name=nvt-na51055
```

```
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
```

```
root=ubi0:rootfs rootfstype=ubifs ubi.fm_autoconvert=1 init=/linuxrc
```

```
boot_mode=0
```

```
bootcmd=loadk;bootm
```

```
bootdelay=3
```

```
cpu=armv7
```

```
dbg=0
```

```
ethact=eth_hik
```

```
...
```

# replacing boot variables

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
HKUS# env set bootdelay 10
```

```
HKUS# env set bootargs "... init=/bin/sh ..."
```

```
HKUS# printenv set bootargs "... init=/bin/sh ..."
```

```
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
```

```
root=ubi0:rootfs rootfstype=ubifs ubi.fm_autoconvert=1 init=/linuxrc
```

```
boot_mode=0
```

```
bootdelay=10
```

# reset the SoC

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
HKUS# env set bootdelay 10
```

```
HKUS# env set bootargs "... init=/bin/sh ..."
```

```
HKUS# printenv set bootargs "... init=/bin/sh ..."
```

```
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
```

```
root=ubi0:rootfs rootfstype=ubifs ubi.fm_autoconvert=1 init=/linuxrc
```

```
boot_mode=0
```

```
bootdelay=10
```

```
HKUS# saveenv
```

```
HKUS# reset
```



# patching the firmware

```
$ diff <(hexdump -C firmware.0) <(hexdump -C firmware)
< 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 6c lonvert=1 init=/1|
< 005000d0 69 6e 75 78 72 63 20 20 20 62 6f 6f 74 5f 6d 6f linuxrc boot_mol

> 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d ?? ?? lonvert=1 init=??|
> 005000d0 ?? ?? ?? ?? ?? ?? 20 20 20 62 6f 6f 74 5f 6d 6f |?????? boot_mol
```

# patching the firmware

```
$ diff <(hexdump -C firmware.0) <(hexdump -C firmware)
< 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 6c lonvert=1 init=/ll
< 005000d0 69 6e 75 78 72 63 20 20 20 62 6f 6f 74 5f 6d 6f linuxrc boot_mol

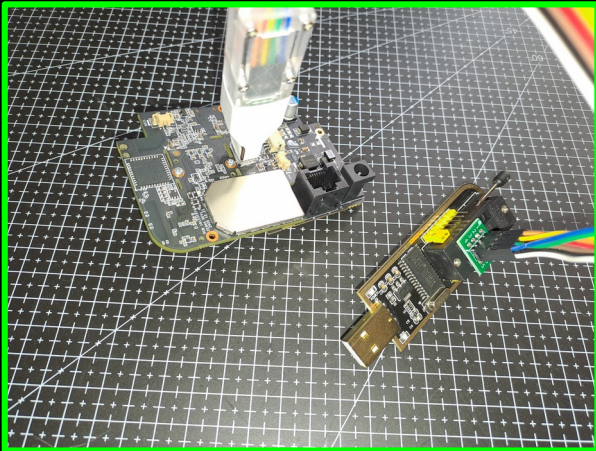
> 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 62 lonvert=1 init=/bl
> 005000d0 69 6e 2f 73 68 20 20 20 20 62 6f 6f 74 5f 6d 6f lin/sh boot_mol
```

# patching the firmware

```
$ diff <(hexdump -C firmware.0) <(hexdump -C firmware)
< 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 6c lonvert=1 init=/l
< 005000d0 69 6e 75 78 72 63 20 20 20 62 6f 6f 74 5f 6d 6f linuxrc boot_mol

> 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 62 lonvert=1 init=/bl
> 005000d0 69 6e 2f 73 68 20 20 20 20 62 6f 6f 74 5f 6d 6f lin/sh boot_mol

$ snander -w firmware.0
```



# unlocked firmware

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
#
```

# questions?

[Repository] <https://github.com/0x0000z3r0/hikvision-re>  
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[LinkedIn] <https://www.linkedin.com/in/hrant-tadevosyan>  
[Discord] 0x0000z3r0 (Z3R0)

?



Github



LinkedIn

